Synapse: Pricing Strategy

Compiled edition, 2026-10-29. This merges the first pricing pass with the second (founder) pass. Where they differ, the second pass wins: three tiers, a launch price, and sponsored seats at RM109. This is for the Hackastone pitch and a post-hackathon launch. Every price is a hypothesis to test, and nothing in the product changes because of this document.

1. The recommendation

> **"Every verified lab in the library is free for every student, forever. You pay only when you want a brand new lab built for your own question: RM12.90 a month for a student, RM15 a year per student for a school."**
>
> The pitch line for judges. It holds because a library lab costs nothing to serve (no model call), while a new lab costs about RM0.24 (measured).

- **Model:** freemium, plus a flat subscription with a monthly allowance of new labs, plus a per-student school licence (section 3).
- **Value metric:** *new labs generated*. Opening, replaying, sharing and exporting library labs is never metered.
- **Tiers:** Free (with teacher features), Plus at RM12.90 a month or RM109 a year, and School at RM15 per student a year (minimum RM1,500). Sponsored seats for B40 students are sold at the Plus yearly price.
- **Launch price:** RM9.90 a month or RM79 a year for the first 90 days, held for founding users (section 8).
- **Goal served:** reach first. Free carries the whole verified library, Bahasa Melayu, offline use and the citations. Trust labels and provenance are never behind a paywall.
- **The honest finding:** student subscriptions alone barely carry the free tier. Two schools at the minimum pay the fixed costs, so the school licence is what makes "free for every student" sustainable.

2. What we know

Cost to serve (our floor, not our basis)

Item	USD	RM (at 4.20)	Source
A library lab (repeat question, or a topic the local router recognises)	0	0	No model call (D-42, P5)
A new lab, measured	0.057	0.24	Paid run of 2026-10-28, 14 questions, per delivered Environment
A new lab, projected range	0.024 to 0.070	0.10 to 0.29	Cost projection, low to pessimistic
A new lab, hard bound with the Free lane	0.159	0.67	Every reply at max tokens; a ceiling, not an expectation
API hosting, Render Standard (2 GB) plus a 1 GB disk	25.25 / month	106 / month	Render, pricing; `render.yaml`
Web hosting, a paid Vercel plan	20 / month	84 / month	Estimate, not checked
Payment gateway fee		1.00 a charge	Estimate (FPX or e-wallet)
Support	0	0	The founder's time, no cash

The planning figures are **RM0.25** per new lab (base), **RM0.35** (stress) and **RM190 a month** fixed (estimate). The lab measurement is 14 questions: a direction, not a rate.

Competitive pricing context (checked 2026-10-29)

Alternative	Price	What's included	Where Synapse sits, and why
Centre tuition, Klang Valley (Malaysia4U)	RM80 to RM300 per subject a month	Classes on a fixed schedule	**Far below.** This is the parent's reference price
GoLearn SPM online	RM49 to RM89 per subject a month	4 live online classes a month	**Far below.** Synapse goes with a teacher, it doesn't replace one
Pandai	Premium Lite RM30 a month, Premium RM96 (its FAQ conflicts)	KSSM practice questions; Premium adds live tuition	**Below.** Synapse covers three sciences, not every subject
ChatGPT Go	RM24 a month (Lowyat, 2026; launched at RM38.99, SoyaCincau)	A general chatbot, not checked against KSSM	**Below.** Plus has to be an easy extra for a student who already pays for a chatbot
Labster	Custom quotes. Older figures: $9 per student a month (2017), about $50 a year (2019)	Virtual science labs for institutions	**Far below** (RM15 is about $3.60), because government schools have little budget. Test RM30 with private and international schools
DELIMa (Tech Wire Asia)	Free to students	The Ministry platform: Google, Microsoft and Apple tools	**Inside it, not against it.** A share link should open from Google Classroom

The free alternatives set the bar for the free tier: Synapse has to be free at first use, or a student never tries it. The paid ones set the ceiling: a parent who pays RM49 a month for one subject sees RM12.90 for three sciences as small.

3. Pricing model and value metric

Model	Fit (1 to 5)	Pros	Cons
Flat subscription	**3**	Parents understand it. Revenue is predictable	Unlimited labs: a heavy user can cost more than they pay at RM0.35 a lab
Usage-based	**2**	Matches cost exactly	Teenagers can't budget per lab. Every question feels like spending
Per-seat	**3**	Right for schools, which buy per student	Wrong for a single student
Freemium	**5**	A library lab costs RM0, so free can be generous. Fits reach first	Needs a paid tier and schools to carry the free users
Credits	**3**	Caps the cost per user	Confusing as the only model. Works better as an allowance inside a plan
One-time purchase	**2**	Easy for a parent	Model cost keeps running after the sale

Recommendation: freemium, plus a flat subscription with a monthly allowance of new labs, plus a per-student school licence. Free is cheap because repeat questions hit the library, the allowance caps the one cost that grows with use, and schools buy yearly and per student.

Candidate value metric	Verdict	Why
New labs generated	**Chosen**	Scales with value and matches our only variable cost. Easy to say ("30 new labs a month"), and hard to game because a repeat question hits the library for free
Labs opened	Rejected	Meters the thing that costs nothing and punishes revision
Per subject	Rejected	Copies tuition centres and splits students who need all three sciences
Flat unlimited	Rejected	One heavy user can cost more than they pay

A student who gets more value asks more of their own questions, hits the free cap and moves to Plus. Each paid lab can also join the shared library after review, so paying users grow what free users get.

4. Tiers

Free

RM0

Every Form 4 and 5 student and teacher

- Every library lab, unlimited (30 verified in English and 30 in Malay today)
- 3 new labs a month
- Hints, predict card, concept check and exam mirror on every lab
- Offline single-file export and share links
- Full citations and trust labels
- Teachers: Present mode, offline chapter packs, and 10 new labs a month

Upgrade trigger: the student asks a 4th question the library can't answer, in a month before an exam.

Plus anchor

RM12.90 / month

or RM109 a year (30% off). Student or parent

- Everything in Free
- 30 new labs a month (about one a day)
- Labs and progress saved across devices*
- Lab report export as a PDF*
- Served first when demand is high during the SPM months

Upgrade trigger: a teacher wants a whole class on Plus, or a parent pays for more than one child.

School

RM15 / student / year

Minimum RM1,500. Annual only, invoiced, sales-led

- Plus for every enrolled student
- A pool of 20 new labs per student a year, shared by the school
- Class codes*
- A shelf of labs chosen by the teachers*
- A termly usage report for the science panel*

Next step: renewal, then a district or state deal.

* Not built. These need accounts, which the PRD lists as a non-goal for v1 (D-11). Everything else exists today: library-first answering, Present mode, chapter packs, offline export, share links, Bahasa Melayu, hints, the exam mirror and per-day limits.

Sponsored seats (reach for B40 students)

A company, a foundation or a state agency buys Plus seats in bulk (50 or more) at **RM109 a seat a year**, the Plus yearly price. There is a local precedent: Air Selangor has sponsored Pandai access for B40 students. The first pass priced these seats at RM79, which loses RM12.50 a seat at full use. The other fix is to keep RM79 and cap sponsored seats at 20 new labs a month.

Why these levels

Perceived value	A month of tuition in one subject: RM49 to RM300
Plus price	RM12.90 a month, all three sciences
Next best paid alternatives	ChatGPT Go RM24, Pandai Premium Lite RM30 a month
Next best free alternative	DELIMa and chatbots: RM0, unverified for KSSM
Cost to serve Plus at full use	RM7.50 base, RM10.50 stress, plus about RM1 in fees

- **RM12.90** is about half of the cheapest paid alternative, and small enough to pay from a student's e-wallet (ASSUMED, test in section 9).
- **RM109 a year** is about RM9 a month and lines up with the SPM school year. Being one payment, it also pays the gateway fee once.
- **RM15 per student a year** is less than a third of one month of one subject at an online tuition centre. A school with 250 science students pays RM3,750 (ASSUMED to fit a PIBG or sponsor budget, test in the teacher interviews). The RM1,500 minimum covers the cost of onboarding a school.
- **Teachers are free** because they are the cheapest route into a classroom. One teacher on Present mode puts Synapse in front of 30 to 40 students for at most RM2.50 a month of new labs.

5. Unit economics

Gross margin per paying unit, per month

Offer	Revenue	Cost, expected (half the cap)	Margin	Cost, full cap	Margin	Cost, full cap, stress	Margin
Free	0	0.25	n/a	0.75	n/a	1.05	n/a
Plus monthly	12.90	4.75	63%	8.50	34%	11.50	11%
Plus yearly and sponsored seat	9.08	3.88	57%	7.63	16%	10.63	-17%
School, per student	1.25	0.21	83%	0.42	67%	0.58	53%
Plus at the launch price	9.90	4.75	52%	8.50	14%	11.50	-16%

All figures are in RM. Free costs 3 labs × 33% × RM0.25. Plus monthly costs 15 × 0.25 + RM1 fee. Plus yearly costs 15 × 0.25 + RM1.50 / 12. School costs 10 labs a year × 0.25 / 12. Margins go negative only if a student uses every lab every month at the stress cost. The spend cap at the LLM seam (`LLM_MONTHLY_CAP_USD`) is the backstop.

Break even

Who pays the fixed RM190 a month	What it takes
Plus only	(190 + 0.25 × free users) / 8.15: 24 subscribers with no free users, or 54 for every 1,000 free users (5.4%, too high to count on)
School only	RM1.04 margin per student a month, so 183 students: **2 schools at the minimum**
Both	Two schools pay the fixed costs. Plus then only pays for the free users: 31 per 1,000 (3.1%)

How big the free cap can be

Free cap	Share of cap used	Cost per free student a month	Conversion that pays for it
3 new labs	33%	RM0.25	**3.0%**
3 new labs	66%	RM0.50	6.1%
5 new labs	33%	RM0.41	5.1%
5 new labs	66%	RM0.83	10.1%

Typical freemium conversion is 2 to 5 percent (an estimate, without a source for this segment). A cap of 3 is about break-even, and 5 is not, so the cap starts at 3. Two things lower the cost of a free student over time. The library answers a growing share of questions at no cost (16 of 53 topics are seeded today). And about half of a plan's input tokens are a stable prefix that prompt caching could make much cheaper, once the saving is measured (D-55).

Target blended ARPU: about RM3.80 per paying student a month (estimate). This assumes 70% school students (RM1.25), 25% Plus yearly (RM9.08) and 5% Plus monthly (RM12.90): 0.88 + 2.27 + 0.65.

6. Pricing psychology

- **Anchoring:** Plus monthly is the anchor, shown next to "one month of tuition in one subject: RM49". RM12.90 for all three sciences then reads as small, and RM49 is a price the parent already pays.
- **Decoy:** Plus monthly is also the decoy for Plus yearly: 12 × RM12.90 = RM154.80 against RM109. A fourth tier as a decoy would confuse 16-year-olds, so there isn't one.
- **Annual framing:** call it the "Form 5 year pass: RM109, about RM9 a month, 3 months free". It maps to the SPM year the student is already counting down, and "3 months free" (RM45.80 saved) lands harder than "30% off".

7. Personas and where each lands

Persona	Need	Willingness to pay	Offer
Form 4 student revising one chapter	See the idea move, check an answer	Near zero; no card	Free
Form 5 student before SPM	Many of their own questions, every day	Low, through Touch 'n Go or a parent	Plus monthly
Parent already paying for tuition	Proof the child is practising; value for money	Moderate; buys a year	Plus yearly
Science teacher	Something to show on the projector tomorrow	Personally none	Free, with teacher features
Head of science panel or PIBG	Coverage of the school, offline on lab PCs	A budget line	School
CSR or foundation	Measurable reach for B40 students	Bulk	Sponsored seats

8. Launch pricing and scale pricing

Launch (first 90 days of Plus): RM9.90 a month or RM79 a year. School pilots are free for one term, in exchange for interviews and usage data. **Because** the library covers only 16 of 53 topics today, the launch has to measure conversion rather than maximise it, and RM9.90 still leaves a 52% expected margin.

Grandfathering: founding Plus users keep RM9.90 or RM79 for as long as they stay subscribed without a break. Pilot schools pay RM10 per student in their first paid year. **Because** these users test the product while it is least finished, and the kept price costs about RM3 a month per user.

Price increases. Move to the scale price when any one of these happens:

1. The library covers 40 of 53 topics (75%), because Plus then buys more.
2. Free to Plus conversion stays at 5% or more for 2 months in a row at the current price.
3. The Van Westendorp survey puts the optimal price above the current price.
4. The measured cost per new lab goes above RM0.35. This raises only the paid allowance, never Free.
5. 80% or more of schools renew.

Give 60 days' notice. Raise prices for new users first and existing users one billing cycle later, and never cut the free library.

9. Research before any price goes live

1. **Van Westendorp** with about 40 Form 4 and 5 students and 20 parents in two schools, on a monthly Plus price. If RM12.90 falls outside the acceptable range, move it before launch.
2. **Gabor-Granger** on the same group at RM5.90, RM9.90, RM12.90 and RM19.90, with parents and students reported separately.
3. **Teacher interviews** (8 to 10): who holds the money (PIBG, the school, the state), the purchase threshold that needs approval, and whether chapter packs matter more than Present mode.
4. **Free cap test**: 3 against 5 new labs a month, assigned by school, never two prices on one page.
5. **Private school price test**: RM30 per student a year, given the Labster reference.
6. **Cost measurement**: a 24-question paid run, which needs the owner's go and a stated amount (rule 7), to replace the 14-question figure.

10. Rollout and measurement

Stage	What happens	Moves on when
0. Hackathon	The public demo (D-75): everything free, behind a $4 monthly cap and per-address and per-day limits. Show this model, not a checkout	Judging is done
1. Pilot	Two or three schools, free for a term, with teachers using Present mode. Run section 9	Library hit rate, labs per student a week and cost per student a week are measured
2. Plus launch	RM9.90 or RM79 for 90 days, held for founding users	Any price-increase trigger in section 8
3. Scale	RM12.90 or RM109 for new users, the School licence, and sponsored seats sold by direct contact	One school renews

Numbers to watch: the library hit rate (it drives everything else), new labs and cost per active student a week (already a deploy-gate item in `checklist.md`), free to Plus conversion (break-even target 3%), monthly churn, the share of Plus users on yearly, school renewal, and students reached per sponsored ringgit.

11. Risks and what must change first

> **This conflicts with the current PRD.** The PRD lists accounts, authentication and payments as non-goals, "ASSUMED permanent for v1" (D-11). Plus, School and saved progress all need accounts. Charging is therefore a v2 decision, and the owner has to reopen D-11 with a new ADR before anything is built. The hackathon pitch can present this model without that change.

- **Textbook legality (A-8).** The corpus belongs to the Ministry of Education, and quoting it in a public product is unconfirmed. Charging money raises the stakes, so confirm the position before taking the first ringgit.
- **Labels are not for sale.** Plus buys more labs, never a better label. A Free lane lab stays `model-checked` whoever paid for it (hard constraints 3 and 11).
- **Children's data.** Accounts for 16 and 17 year olds bring in the Personal Data Protection Act 2010 and parental consent. That build cost isn't priced here.
- **The deploy gate.** The public demo runs with the gate waived (D-75). A paid launch needs the gate in `checklist.md` fully met.
- **Thin cost evidence.** One 14-question run. The stress case (RM0.35) and the spend cap cover the gap until the 24-question run.
- **Free competitors.** If DELIMa or a chatbot adds cited, checked labs, the free tier has to stay ahead on KSSM coverage and offline use, not on price.
- **Exchange rate.** Costs are in USD and prices in RM. A 10% move in the ringgit changes the expected Plus margin by about RM0.40.

12. Validation checklist

☑ Clear value metric: new labs generated
☑ Distinct personas, each mapped to one offer
☐ Research-backed price range (section 9 not yet run)
☑ Conversion-safe entry tier: free, no card, the whole library

☑ Expansion path: Free to Plus to School, with teachers as the route into schools
☑ Institutional buyers handled explicitly: School licence and sponsored seats
☑ Launch price, grandfathering and price-increase triggers set
☐ D-11 reopened by the owner before anything is built

Sources and assumptions

- Cost: `refdocs/evidence/2026-10-28-w16-paid-measurement.md` (measured) and `refdocs/evidence/2026-10-16-p3-cost-projection.md` (projected). Hosting: `render.yaml` and Render's own pages, linked above.
- Competitor prices are linked in section 2, all read on 2026-10-29. Pandai's pages conflict, the ChatGPT Go price comes from news reports rather than OpenAI's page, and Labster's figures are from 2017 and 2019. Check each one before quoting it.
- ASSUMED: RM4.20 per USD; a payment fee of about RM1 a charge; Vercel at $20 a month; expected use at half the cap; 2 to 5% typical freemium conversion; the user mix behind the blended ARPU.
- Compiled from `refdocs/business/2026-10-29-pricing-strategy.pdf` (first pass) and `founder/pricing-strategy.md` (second pass). Those files are unchanged.